

Multiple Choice (10 marks)

1. Which of the following is least likely to reduce bias in a sample survey?
 - (a) Following up with those who did not respond to the survey the first time
 - (b) Asking questions in a neutral manner to avoid influencing the responses
 - (c) Using stratified random sampling rather than simple random sampling
 - (d) Selecting samples randomly
2. How many years are there in a decade?
 - (a) 5
 - (b) 10
 - (c) 15
 - (d) 20
3. The domain of the function $h(x) = \sqrt{25 - x^2} + \sqrt{-(x - 2)}$ is an interval of what width?
 - (a) 7
 - (b) 5
 - (c) 2
 - (d) 25
4. Members of the Hiking Club hiked for 2.4 hours covering 2.5 miles each hour. How far did the members hike
 - (a) 4.9 miles
 - (b) 5.8 miles
 - (c) 6.0 miles
 - (d) 6.3 miles
5. John has five children. What is the probability that at least half of them are girls? (We can assume a boy is equally likely to be born as is a girl, and vice-versa.)
 - (a) $\frac{1}{2}$
 - (b) $\frac{1}{4}$
 - (c) $\frac{1}{8}$
 - (d) $\frac{3}{5}$

True / False (10 marks)

Answer True or False

6. True or False: Dividing 942 by 3 results in 214, which is the correct answer.
7. True or False: The sum of the center coordinates and the radius of the circle C is -9.
8. True or False: A 6-lb bag of cat food costs 2.82 when priced at 0.47 per pound.
9. True or False: The equation $y - 36 = 13$ correctly represents 36 less than a number y being equal to 13.
10. True or False: It takes the machine 25 minutes to fill 500 bottles of soda at the given rate.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Henry's Hamburger Heaven offers its hamburgers with the following condiments: ketchup, mustard, mayonnaise, tomato, lettuce, pickles, cheese, and onions. A customer can choose one, two, or three meat patties, and any collection of condiments. How many different kinds of hamburgers can be ordered?
12. Discuss how you would translate the phrase "3 fewer than a number, p" into a mathematical expression. Explain your reasoning and the steps taken to arrive at your answer.

End of Paper